

YASH RAJ PANDEY

Gainesville, FL | +1 (786) 874-7467 | yashraj.pandeyyy@gmail.com | [LinkedIn](#) | [GitHub](#)

SUMMARY

Full-Stack Software Engineer building production platforms that solve complex domain problems at scale. Experience spans scientific research (LIMS for 30+ researchers managing 150K+ records), legal tech, and real-time collaboration systems. Strong in React/TypeScript/Flutter frontends, Django/Python backends, and cloud infrastructure, with a focus on scalable architecture, automation, performance optimization, and seamless user experiences.

TECHNICAL SKILLS

Languages: JavaScript, TypeScript, Python, Dart, R, SQL

Frontend: React.js, Flutter, Material-UI, Plotly, Recharts, Nivo, D3.js, IGV.js, JBrowse

Backend: Django, Django REST Framework, FastAPI, Node.js, PostgreSQL, Redis

Cloud & DevOps: GCP (GKE), AWS, Docker, Kubernetes, Terraform, GitHub Actions

Data & Tools: R (rmarkdown), Statistical Analysis, OpenAI API, RAG, FAISS, BLAST, MAFFT

PROFESSIONAL EXPERIENCE

Software Engineer | Institute of Food & Agricultural Sciences

Mar 2025 – Present

- Designed and delivered a full-stack LIMS platform for **30+ researchers across 5 labs**, developing a **React** frontend with genome visualization (**IGV.js**, **JBrowse**) and analytics dashboards (**Plotly**, **Nivo**) backed by a **Django REST** backend.
- Engineered comprehensive REST APIs with unified search and complex relational queries; integrated external bioinformatics tools (**BLAST**, **MAFFT**, **InterPro**) and enhanced **PostgreSQL** query performance and caching for concurrent access by 30+ users.
- Automated analytical report generation pipelines using **R Markdown**, standardizing outputs, reducing manual processing from **2–3 hours to 15–20 minutes**, and enabling reliable quarterly data releases.
- Deployed the platform on **GCP** using **Kubernetes** and **Terraform**; created **Docker** multi-stage builds, introduced JWT authentication and CI/CD pipelines, and improved frontend load time (8s → 3s), increasing daily users by **40%**.

Graduate Assistant | Systems Operations | RecSports

Jul 2024 – Dec 2024

- Championed operational excellence by vigilantly enforcing protocols, orchestrating facility workflows, and monitoring equipment, achieving a **100% service availability record**, and uninterrupted access for 500+ daily users.
- Conducted real-time issue detection, incident response, and operational reporting, resolving equipment and system anomalies quickly to minimize downtime and ensure uninterrupted daily operations.

Software Developer Intern | Hackdev Technology Pvt. Ltd.

Jan 2022 – Jun 2022

- Conceived a **Flutter** mobile solution connecting scam victims with relevant legal professionals; designed intelligent intake flows which achieved a 80% intake flow completion rate with new users.
- Designed and deployed a case assignment system matching users with specialized lawyers; integrated **Firebase Realtime Database** for case management, availability tracking, and consistent real-time updates.
- Enhanced real-time data synchronization using query profiling, batch operations, and **Redis** caching, reducing sync latency from **4–5s to under 2s** during peak user activity.
- Transitioned manual deployment processes to fully automated **CI/CD pipelines** via **GitHub Actions**, achieving a 90% reduction in deployment duration, which helped the company save approximately 5 hours per week..

PROJECTS

CodePilot AI | Developer Productivity Assistant

Aug 2024 – Dec 2024

React, TypeScript, Python, FastAPI, OpenAI API, FAISS, Pinecone, GitHub Webhooks, GCP

- Architected an AI-powered developer assistant using a **RAG** architecture, combining **OpenAI API**, **FAISS**, and **Pinecone** to analyze large codebases, answer contextual queries, and surface PR issues via **GitHub webhooks**.
- Delivered interactive dashboards in **React** to visualize code health metrics and semantic insights; enabled vector-based search across **10K+ files** with optimized indexing, achieving response times under **500ms**.

SynapseFlow | Real-Time Collaborative Analytics Platform

Jan 2023 – Jul 2024

React, TypeScript, FastAPI, PostgreSQL, Redis, WebSockets, Kafka, Kubernetes, GCP

- Designed and launched a real-time collaborative analytics platform using **React** and **TypeScript**, enabling optimistic UI updates, WebSocket-based synchronization, and conflict resolution for simultaneous multi-user interactions with sub-second query latency.
- Scaled backend services using **FastAPI**, **PostgreSQL**, and **Redis**, leveraging **Kafka** for event-driven updates and horizontal scaling via **Kubernetes**; load-tested with **100+ concurrent users**.

EDUCATION

University of Florida

Jan 2023 – Dec 2024

M.S. in Computer & Information Science & Engineering (GPA: 3.8/4.0)

Jaypee University of Engineering and Technology

Jul 2019 – May 2023

B.Tech. in Computer Science & Engineering (GPA: 9.1/10.0)